

headnote.

PRODUCT REQUIREMENTS DOCUMENT • V0.1

WhatsApp Bot — PRD

End-to-end legal research over WhatsApp. Verified citations and a downloadable PDF memo — delivered inside the chat thread
Indian advocates already live in.

Owner	Ayush (xyush14)	Last updated	2026-06-15
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Headnote WhatsApp Bot — Product Requirements Document

End-to-end research assistant for Indian advocates over WhatsApp. Lawyer sends a natural-language legal question, bot returns verifiable citations plus a downloadable PDF memo — all inside the chat thread.

This PRD is the contract for v1. It is written to be implementable by a solo founder + one contractor in **3–4 focused weeks** on a **₹0–₹2,500 per month** operating budget at launch.

• 1. Why this exists

Indian legal professionals live on WhatsApp. Drafting a brief, asking a junior to "find me a case on X", forwarding judgments — all of it happens in DMs and group chats. The friction of opening a browser, logging into a research portal, formulating a search, and copying back results means that **most spot-research questions get a guessed answer instead of a real one.**

Headnote already has the research engine — IK→official-source re-anchored citations, official judgment corpus, drafter, paid tier. What it lacks is the channel that meets lawyers where they already are.

A WhatsApp bot collapses the entire "I need a case on this" loop into one DM:

- **Lawyer:** s.138 NI Act recent SC on territorial jurisdiction
- **Bot** (under 30s): 5 verified citations + 1-line ratios + PDF memo attached, send-forwardable to the junior.

This is the single highest-leverage distribution channel for Headnote. The competitive observation that finalized the call: **NyayAssist has a WhatsApp bot, but it's for file management — not research.** The research-via-WA slot is open.

• 2. Goals and non-goals

V1 GOALS

- **G1.** A working WhatsApp number that any Indian lawyer can DM and receive a citation-backed research response within 60 seconds.
- **G2.** Phone-number-based account linkage that recognizes existing paid Headnote users and removes their quota.
- **G3.** A free-tier funnel that lets new lawyers try 3 queries/day and converts them to paid via a link to headnote.in.

- **G4.** Zero net-new LLM spend per query beyond what existing Headnote research costs (i.e., reuse existing DeepSeek/Groq path).
- **G5.** Operate at **₹0 marginal infra cost** for the first 1,000 service conversations per month (Meta Cloud API free tier).

V1 NON-GOALS (DEFERRED TO V2+)

- Voice-note transcription (Whisper/Groq integration deferred)
- Hindi / regional language input/output (English-only at v1)
- Document upload (lawyer sending a PDF and asking "find similar cases")
- Multi-turn long-form drafting inside WhatsApp
- Group-chat support (DM only at v1)
- Push notifications / proactive case alerts
- Calendar / hearing reminders
- Payments inside WhatsApp (link out to headnote.in instead)

EXPLICIT OUT-OF-SCOPE

- Replacing the Headnote web app — the bot is a channel, not a fork
- Replacing email — PDF memo is delivered in WA AND emailed for archival
- Hosting any LLM ourselves — all inference stays on existing providers

• 3. Target users

PRIMARY PERSONA — SOLO ADVOCATE, 2-10 YEARS PQE

- Practices in district / High Court
- Owns the matter end to end; researches for self
- Lives in WhatsApp; checks email twice a day
- Currently uses: free Indian Kanoon, sometimes SCC Online via friend's login
- Pain: spends 30-90 minutes per matter looking up precedent
- Willingness to pay: ₹500-₹1,500 / month for something that saves the hour
- This user is **the dominant Headnote ICP** and is who the bot is built for

SECONDARY — JUNIOR ASSOCIATE IN A 5-20 LAWYER FIRM

- Researches on partner's behalf, gets WhatsApp pings at 11pm
- Often the one who'll *forward* the bot's output to seniors
- Doesn't pay directly but is the strongest distribution multiplier

TERTIARY – LAW STUDENT PREPARING MOOTS

- High-frequency low-value queries
- Stays on free tier indefinitely; useful for word-of-mouth and long-tail conversion when they start practice

• 4. User stories (v1)

#	AS A...	I WANT TO...	SO THAT...
US1	new lawyer	DM the bot and get a useful answer without signing up	I can evaluate it before deciding to subscribe
US2	existing paid Headnote user	link my WhatsApp number once and forget about it	I get unlimited research from any device
US3	lawyer in a hurry	type a 1-line question and get citations in under a minute	I can answer the partner's question on the road
US4	lawyer who got a useful result	forward the response to a colleague	the colleague sees Headnote and tries it
US5	lawyer who hit free quota	upgrade in two taps	I don't lose my flow
US6	lawyer with a follow-up	refine my last query without re-typing context	conversation feels natural
US7	lawyer who wants the full memo	get a PDF with full ratios, paragraph quotes, and links	I can use it in my brief
US8	lawyer worried about privacy	trust that the bot doesn't leak my query	I keep using it on sensitive matters

• 5. Conversational UX

FIRST-TIME DM (COLD CONTACT)

[user] Hi

[bot] Welcome to Headnote – citation-checked legal research on WhatsApp.

I find verified Indian case law and statutes. Try a question like: "Section 138 NI Act recent SC on territorial jurisdiction" or "doctrine of part performance latest HC view".

You get 3 free queries per day. Existing Headnote subscribers get unlimited – just send LINK to connect your account.

RESEARCH QUERY (THE CORE LOOP)

[user] section 138 NI act recent SC on territorial jurisdiction

[bot] 🔍 Searching the SC corpus...

[bot] Found 5 SC judgments on Section 138 NI Act / territorial jurisdiction (last 5 years):

1. *Dashrath Rupsingh Rathod v. State of Maharashtra*
(2014) 9 SCC 129
→ Cheque return triggers cause of action; complaint lies where drawee bank is located.
2. *Bridgestone India v. Inderpal Singh*
(2016) 2 SCC 75
→ Clarifies post-Dashrath position after 2015 Amendment; place of payee's bank now central.
3. *K. Bhaskaran v. Sankaran Vaidhyan Balan*
(1999) 7 SCC 510
→ Five-act test, foundational pre-amendment frame.
4. *Yogendra Pratap Singh v. Savitri Pandey*
(2014) 10 SCC 713
→ Limitation interplay with cause of action.
5. *M/s Yashshree Talkies v. State of Maharashtra*
(2023) – pending judgment reference

📎 Full memo sent: 4-page PDF with quotes, paragraph numbers, and direct links to indiankanoon.org + the SCI source.

Queries left today: 2/3 free.

Reply REFINER <new angle> to narrow this search.

Reply UPGRADE to remove daily limit (₹999/mo).

ACCOUNT LINKAGE (EXISTING PAID USER)

[user] LINK

[bot] Got it. Reply with the email or phone you use on headnote.in and I'll send a 4-digit code to confirm.

[user] kshitij@loopai.com

[bot] Code sent. Check your headnote.in inbox or SMS.

[user] 7421

[bot]  Linked. You're on the Advocate plan – unlimited queries on WhatsApp from now on. Send a research question to begin.

QUOTA EXHAUSTION

[user] (4th query of the day, free tier)

[bot] You've used your 3 free queries today. They reset at midnight IST.

Headnote subscribers get unlimited WhatsApp research, full PDF memos, and access to the web app:

→ Subscribe: <https://headnote.in/upgrade>

Already subscribed? Reply LINK to connect your account.

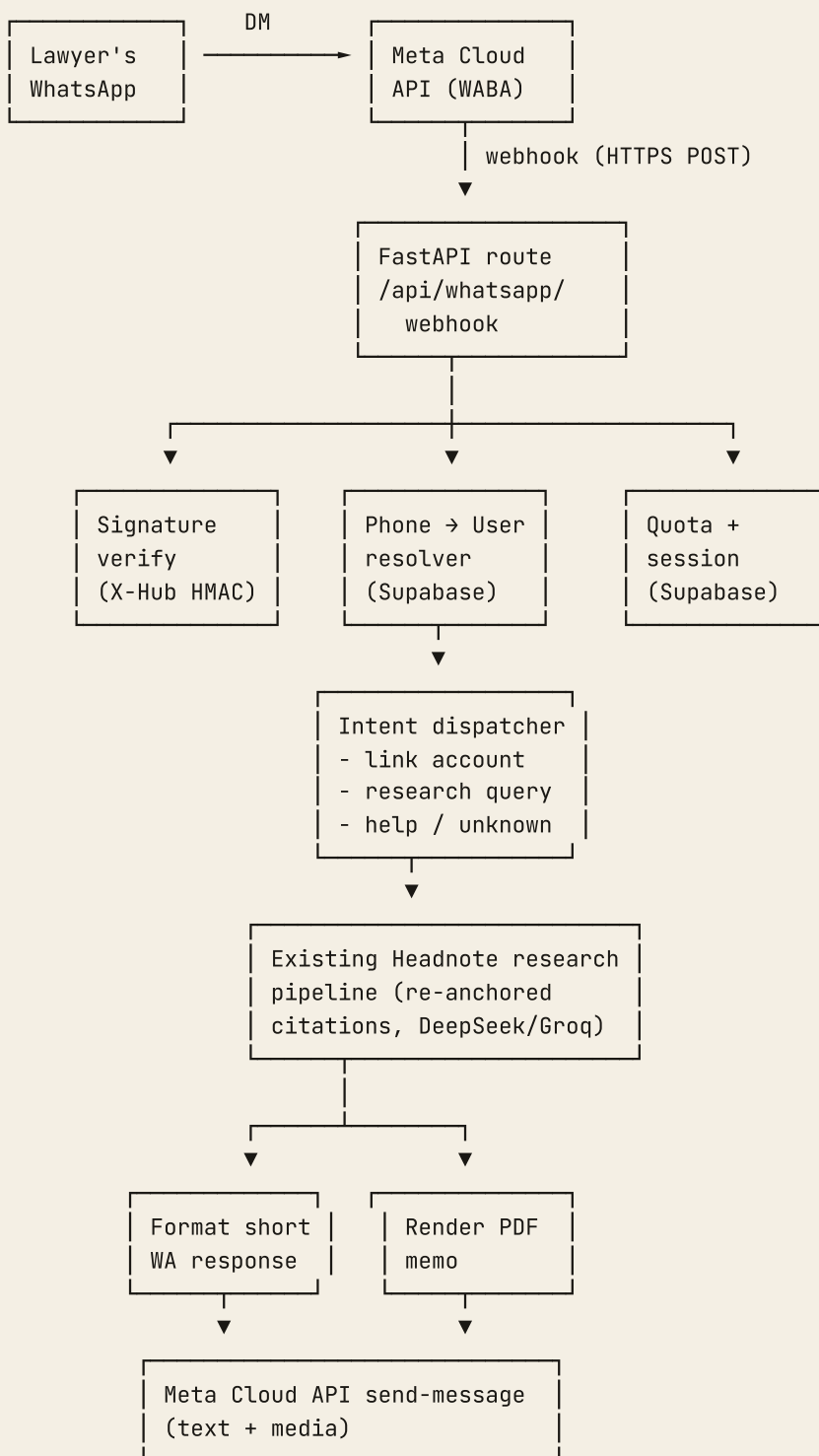
OUT-OF-WINDOW RE-ENGAGEMENT (24H+ LATER)

The 24h customer-service window is closed. Bot cannot send free-form text. It can send an *approved utility template* like:

[bot] (utility template) Your daily Headnote queries have reset.
Reply with any legal research question to begin. – Headnote

We will register **one** utility template at v1. Marketing pushes are explicitly out of scope (we don't want to be the kind of bot lawyers mute).

• 6. Technical architecture



WHY META CLOUD API DIRECT (NOT A BSP)

We compared 5 paths. Decision: **Meta Cloud API direct**, because at tight budget the BSP layer is a tax we don't need yet.

PROVIDER	MONTHLY MINIMUM	INDIA BILLING	DX	VERDICT
Meta Cloud API direct	₹0	✓ (per-conversation, INR via Meta)	OK — write own webhook	v1 pick
AiSensy	₹999 + convo fees	✓ INR	UI for templates	Reconsider at scale
Interakt	₹999+	✓ INR	Decent UI	Same as AiSensy
Gupshup	Pay-as-you-go	✓ INR	Heavier integration	Enterprise sizing
Wati	\$49+ USD/mo	✗ USD	Best DX of all	Skip — USD pricing
Twilio	\$0.005/msg + WA fees	✗ USD	Cleanest API	Skip — no India edge

A BSP is a hosted UI for managing templates, analytics, and broadcasts. We don't need any of that yet. The Meta Cloud API gives us **1,000 free service conversations per month** and per-conversation pricing after. Webhook handling is one FastAPI route — we already write FastAPI.

We can move to a BSP at any time later (DNS-level change essentially).

CODE SURFACE – WHAT WE ADD

```
headnote/
├── api/
│   └── whatsapp.py          # NEW – webhook, signature verify, dispatch
├── whatsapp/               # NEW package
│   ├── __init__.py
│   ├── client.py           # Meta Cloud API send-message wrapper
│   ├── handlers.py         # intent → handler map
│   ├── session.py          # conversation state, quota, linkage
│   ├── templates.py        # registered template names + senders
│   └── formatters.py        # WA-text + PDF memo rendering
└── migrations/
    └── 000X_whatsapp.sql    # users.wa_phone, wa_messages log, wa_quota
```

REUSED (ZERO NET-NEW CODE)

- Research pipeline: `headnote/situation_pipeline.py` and `headnote/api/assist.py`
- LLM dispatch: existing DeepSeek/Groq path in `headnote/llm/`
- PDF generation: `headnote_caselaw_pdf.py` pattern (adapt to memo format)
- Auth: `headnote/api/auth_otplless.py` — phone-OTP already wired to Supabase. We piggyback the same flow for WhatsApp linking.
- DB: existing Supabase Postgres

• 7. Data model (delta only)

Three additions to existing schema. Migration file `migrations/00X_whatsapp.sql` :

```
-- 1. Add WhatsApp phone to user profile
ALTER TABLE users
  ADD COLUMN wa_phone TEXT UNIQUE,          -- E.164, e.g. +919876543210
  ADD COLUMN wa_linked_at TIMESTAMPTZ;

CREATE INDEX users_wa_phone_idx ON users(wa_phone);

-- 2. Message log (debugging, abuse detection, analytics)
CREATE TABLE wa_messages (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  wa_phone    TEXT NOT NULL,
  direction   TEXT NOT NULL CHECK (direction IN ('in', 'out')),
  msg_type    TEXT NOT NULL,  -- text, document, template, system
  body        TEXT,           -- redacted to first 500 chars
  meta_msg_id TEXT,           -- id from Meta payload
  user_id     UUID REFERENCES users(id),
  created_at  TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX wa_messages_phone_created_idx
  ON wa_messages(wa_phone, created_at DESC);

-- 3. Per-day quota counter for free users
CREATE TABLE wa_quota (
  wa_phone    TEXT NOT NULL,
  day         DATE NOT NULL,      -- IST day (Asia/Kolkata)
  count       INT NOT NULL DEFAULT 0,
  PRIMARY KEY (wa_phone, day)
);
```

All other state — research history, memo PDFs, payments — lives in existing tables. No new big tables.

• 8. Functional requirements

F1. WEBHOOK INGESTION

- **F1.1** Endpoint: `POST /api/whatsapp/webhook` . Public. Verifies Meta's `X-Hub-Signature-256` HMAC using `WA_APP_SECRET` . Reject 403 on mismatch.
- **F1.2** Endpoint: `GET /api/whatsapp/webhook` . Used by Meta for initial verification handshake. Echoes `hub.challenge` if `hub.verify_token == WA_VERIFY_TOKEN` .
- **F1.3** All incoming messages logged to `wa_messages` (direction = 'in') before dispatch.
- **F1.4** Bot replies within 2s "ack" (typing indicator + status message for queries > 5s).

- **F1.5** Idempotency: dedupe by `meta_msg_id` — Meta retries on failure, we must not double-process.

F2. INTENT DETECTION

- **F2.1** Simple keyword router at v1 (not LLM-based). Patterns:
 - First-time user → onboarding flow
 - `LINK` (case-insensitive) → account linkage
 - `UPGRADE` → send upgrade link
 - `HELP` → send help text
 - `REFINE <text>` → refinement of last query (uses session memory)
 - Anything else → treat as research query
- **F2.2** Research queries go to existing `assist.py` pipeline.
- **F2.3** Unknown / unparseable input → fallback help message, do not consume quota.

F3. PHONE-TO-ACCOUNT LINKAGE

- **F3.1** First message from a new phone creates a `wa_users` shadow record (not a full user). They're a "trial" user.
- **F3.2** When user sends `LINK`, bot asks for email/phone used on headnote.in.
- **F3.3** Bot reuses existing OTPless flow: triggers an OTP send to that email/phone via the existing `auth_otpless.py` machinery.
- **F3.4** User replies with OTP in WhatsApp. Bot verifies via the existing OTPless verify endpoint, then writes `users.wa_phone = <sender_wa_phone>`.
- **F3.5** From then on, that WA number is treated as the linked user with full plan entitlements.
- **F3.6** Re-linking same WA phone to a different user requires manual admin action (security — prevents account hijack via SIM swap).

F4. QUOTA AND ENTITLEMENTS

- **F4.1** Linked paid users: unlimited research queries; no counter shown.
- **F4.2** Trial / unlinked phones: 3 research queries per IST day. Counter shown after every response: "Queries left today: N/3 free".
- **F4.3** On 4th query, return upgrade CTA. Do not run the research.
- **F4.4** Help / link / unknown messages do NOT consume quota.
- **F4.5** Quota reset job: nothing — natural day rollover via the `day` column in `wa_quota`.

F5. RESEARCH EXECUTION

- **F5.1** Call existing situation pipeline with the user's text as the natural-language query.
- **F5.2** Use existing DeepSeek V3 default (V3 for fast path). Per user memory: **never Claude** at the bot serving path.
- **F5.3** Use Groq Llama 3.x as fallback if DeepSeek errors.

- **F5.4** Result must include: 3–5 citations with court, year, cite ref, 1-line ratio, and a verified link (re-anchored to official source per existing Headnote design).
- **F5.5** Generate a 1–3 page PDF memo using `headnote_caseLaw_pdf.py` format. Send as document attachment in same WA reply thread.

F6. OUTPUT FORMATTING

- **F6.1** WhatsApp text response: under 1,000 characters when possible. Use *italics* for case names, plain text otherwise (no markdown beyond what WA renders).
- **F6.2** Citation block format: `N. *Case Name v. Other Party* Citation (Year) → One-line ratio / holding`
- **F6.3** PDF memo: 1–3 pages, Headnote branding, full quotes with paragraph numbers, link to source.
- **F6.4** PDF filename: `headnote_<short-query-slug>_<YYYYMMDD>.pdf` .

F7. CONVERSATION MEMORY

- **F7.1** v1 keeps the **last query + last result** per WA phone in Supabase (column on `wa_quota` or a lightweight table).
- **F7.2** `REFINE <text>` rewrites the previous query incorporating the new angle, runs again. Consumes 1 quota.
- **F7.3** No multi-turn dialogue at v1 beyond REFINE.

F8. COMPLIANCE BASICS

- **F8.1** Privacy notice in onboarding message + linked at `headnote.in/privacy` .
- **F8.2** `STOP` keyword unsubscribes (no further proactive messages).
- **F8.3** Message log retains payloads for 90 days, then purged automatically. Aggregate stats retained.
- **F8.4** No sharing of query content with any third party beyond the LLM provider used to fulfill the request (DeepSeek / Groq).

• 9. Non-functional requirements

ID	REQUIREMENT	TARGET
NF1	P50 research response latency	< 30s
NF2	P95 research response latency	< 60s
NF3	Citation correctness	≥ 95% verifiable to live source
NF4	Uptime	≥ 99% (matches existing Render hobby tier)
NF5	Webhook ack latency	< 2s

ID	REQUIREMENT	TARGET
NF6	LLM cost per query	< ₹0.10
NF7	Infra cost (excluding LLM)	₹0 to first 1,000 conv/mo

• 10. Pricing & cost model

This is the section to look at twice. Tight-budget posture throughout.

10.1 ONE-TIME SETUP COSTS

ITEM	COST	NOTES
WhatsApp Business Number (new SIM)	₹500	Or reuse an existing non-WA-registered number. Cannot be a number currently on WhatsApp consumer app.
Facebook Business Verification	₹0	Document upload; 3–5 business days
WABA Display Name approval	₹0	Submitted via Meta Business Suite; 24–48h
Template approval (1 utility template)	₹0	24–48h
Development time (solo or 1 contractor)	₹0–₹40,000	If outsourced: ~₹30–40k for 3 weeks
Total cash setup	₹500–₹40,500	Lower bound if you build it; upper if you outsource

10.2 RECURRING INFRASTRUCTURE

Meta Cloud API conversation pricing (India, as of late 2025 — verify on launch). Per-conversation, free first 1,000/month.

CONVERSATION TYPE	PER-CONVERSATION COST (INDIA)	FREE TIER
Service (user-initiated, 24h window)	~₹0.30	1,000 / month
Utility template	~₹0.16	shared cap
Marketing template	~₹0.88	shared cap
Authentication template	~₹0.12	shared cap

⚠️ Verify exact current rates at <https://developers.facebook.com/docs/whatsapp/pricing> — Meta has changed pricing twice in the past two years. The PRD assumes "service conversations are free up to 1,000/month" which has been stable since 2024.

At launch scale (≤ 1,000 service conversations / month):

ITEM	MONTHLY COST (INR)
Meta Cloud API conversations	₹0
Render web hosting	₹0 (existing free tier)
Supabase	₹0 (existing free tier)
LLM (DeepSeek V3, ~500 queries/mo @ ₹0.05)	₹25
LLM (DeepSeek R1 deep, ~50 queries/mo @ ₹0.50)	₹25
Groq fallback	₹0 (free tier sufficient)
Domain / SSL	₹0 (existing)
PDF storage (~50MB/mo)	₹0 (Supabase storage free tier)
Estimated launch month-1 spend	₹50–₹200

At 5,000 conversations / month (post-PMF scale):

ITEM	MONTHLY COST (INR)
Meta Cloud API (4,000 paid service conv @ ₹0.30)	₹1,200
Meta Cloud API (utility template, 500 @ ₹0.16)	₹80
Render upgrade (Starter tier)	₹600 (~\$7)
Supabase Pro (if storage exceeded)	₹2,100 (~\$25)
LLM (5,000 queries, mix of V3 + R1)	₹500–₹1,500
PDF storage / CDN	₹100
Estimated scale spend	₹4,500–₹5,500

10.3 LLM COST DETAIL (PER-QUERY WORKED EXAMPLE)

Reuses existing Headnote pipeline. Per user memory: never Claude.

STAGE	PROVIDER	TOKENS (IN+OUT)	COST/QUERY
Decompose query	DeepSeek V3	~600	₹0.005

STAGE	PROVIDER	TOKENS (IN+OUT)	COST/QUERY
Retrieve (local SQLite, no LLM)	—	—	₹0.000
Rank / cluster	DeepSeek V3	~2,000	₹0.015
Synthesize citation block	DeepSeek V3	~1,500	₹0.012
Render memo prose	DeepSeek V3	~3,000	₹0.025
Total per fast-path query		~₹0.06	

Deep / R1 queries (~10% of traffic if user is doing serious research): ~₹0.40 per query. Blended estimate: **₹0.08–₹0.12 per query.**

10.4 REVENUE MODEL (WHAT THE BOT FUNNELS INTO)

The bot is a **funnel to existing Headnote paid plans**. No new SKU needed at v1.

Suggested funnel math at break-even:

- 100 lawyers DM the bot in month 1 (organic + IG-driven)
- 50 use up to free quota (50% activation)
- 20 link an existing account (good — existing users get value)
- 5 convert from cold to paid (5% cold conversion) at ₹999/mo
- Month-1 revenue lift: **₹4,995**
- Month-1 infra cost: **₹50–₹200**
- Net: **+₹4,795 in month 1** at break-even-light assumption

The economics work because Headnote has an existing paid tier. The bot is a distribution multiplier, not a cost center.

10.5 WHEN TO SPEND MORE

Triggers to upgrade infra:

TRIGGER	WHAT TO UPGRADE	NEW MONTHLY COST
> 800 conversations / mo	Nothing yet — watch free tier	—
> 1,200 conversations / mo	Accept Meta paid tier (auto)	+ ₹500–₹1,500
P95 latency > 90s	Render Starter (\$7/mo)	+ ₹600
> 1GB PDF storage	Move PDFs to S3 (Backblaze B2 cheaper: \$0.005/GB)	+ ₹50
> 50 simultaneous DM users	Consider BSP (AiSensy) for queue / templates	+ ₹999

Do not preemptively upgrade. Watch the metrics in §13 and respond.

• 11. Security & compliance

11.1 WHATSAPP / META REQUIREMENTS

- **WABA verification** — Facebook Business Manager → submit incorporation docs (Headnote LLP/Pvt Ltd PAN, GST cert, address proof). Required before any bot can go live.
- **Display name approval** — "Headnote" or similar. Cannot use trademarked names without permission.
- **Phone number** — dedicated, never registered on WhatsApp consumer app. Either a fresh SIM or a number you've de-registered first via <https://faq.whatsapp.com/>.

11.2 DATA HANDLING

- All inbound messages stored in `wa_messages` for 90 days, then purged. Aggregate stats (count by day, query category) retained indefinitely.
- No query text or response logged with any third party beyond the LLM provider serving the request.
- **STOP** keyword: writes `users.wa_unsubscribed_at`, halts all outbound messages, retains data per retention policy.
- LLM provider data policies: DeepSeek and Groq both offer no-training-on-customer-data tiers; confirm setting before launch.

11.3 WEBHOOK SECURITY

- HMAC verify on every inbound webhook (`X-Hub-Signature-256`).
- IP allowlist of Meta's published webhook IP ranges (optional but recommended at scale).
- Per-phone rate limit: 30 messages / minute, 200 / hour. Above threshold → log + ignore (don't bounce, since bouncing burns convo).

11.4 DPDPA POSTURE (INDIA, 2023 ACT)

Not full audit at v1 — a clear privacy page at headnote.in/privacy covering:

- Categories of personal data collected (phone, query text)
- Purpose (legal research delivery)
- Retention (90 days for messages, indefinitely for aggregates)
- User rights (access, deletion via STOP + email request)
- Cross-border transfer (LLM providers — disclose explicitly)
- Grievance officer (your name + email)

This is sufficient for solo founder posture. Full SOC2 / ISO is a post-Series-A problem.

• 12. Metrics and KPIs

12.1 NORTH-STAR

Weekly Active Lawyers on WhatsApp — distinct WA phones that sent ≥ 1 research query in the last 7 days. Tracks reach + engagement in one number.

12.2 FUNNEL

STAGE	METRIC	TARGET MONTH 1	TARGET MONTH 3
Acquisition	Unique WA phones DMing	100	500
Activation	% that send ≥ 1 research query	70%	80%
Linkage	% that send LINK	20%	30%
Retention	Day-7 return rate (any message)	25%	40%
Conversion	% cold $\rightarrow$ paid in 30 days	3%	6%
Revenue	New paid users / mo via WA	3	30

12.3 QUALITY

METRIC	TARGET
Citation correctness (manual sample of 50/mo)	$\geq 95\%$ verifiable
P50 latency	$< 30s$
P95 latency	$< 60s$
Failed responses (errors / total)	$< 2\%$
Spam / abuse messages	$< 5\%$ of inbound

12.4 COST

METRIC	TARGET
Cost per query (LLM + infra)	$< ₹0.15$
Cost per acquired paid user	$< ₹500$

12.5 WHERE THE METRICS LIVE

- Counters: increment in `wa_messages` aggregation views
- Funnel: simple weekly SQL pulled in Supabase studio
- Latency: log on each webhook handler, daily P50/P95 via SQL

- Quality: manual review queue — 1 hour/week to spot-check 50 queries

No external analytics product at v1. Supabase + a Sunday SQL hour covers it.

• 13. Rollout plan

PHASE 0 — PRE-BUILD (WEEK 1)

- ☐ Get a fresh SIM card not registered on WhatsApp
- ☐ Create Facebook Business account; submit business verification documents (PAN, GST, address proof)
- ☐ Create a WhatsApp Business App in Meta Business Suite
- ☐ Submit Display Name "Headnote" for approval
- ☐ Set up webhook URL placeholder on Render
- ☐ Note: FB verification is the **long pole** — 3–7 business days. Start immediately.

Cost: ₹500 (SIM) + free

PHASE 1 — BARE-LOOP BOT (WEEK 2)

- ☐ Webhook GET handshake + POST signature verify
- ☐ Echo bot: any inbound message → "Got it: " outbound
- ☐ Logging to `wa_messages`
- ☐ Deploy to Render
- ☐ Test with personal WA → bot number

Acceptance: a message sent to the bot number gets echoed back in under 2s.

PHASE 2 — RESEARCH PIPELINE INTEGRATION (WEEK 3)

- ☐ Wire research pipeline → format response + generate PDF
- ☐ Send text + PDF document attachment via Meta API
- ☐ First-time onboarding message
- ☐ Help / unknown / STOP keyword handling
- ☐ Privacy page live at headnote.in/privacy

Acceptance: 10 representative queries (you + senior advocate) return correct citations + PDF attachment within 60s. No critical bugs in formatting.

PHASE 3 — LINKAGE + QUOTA (WEEK 4)

- ☐ LINK flow reusing OTPless
- ☐ Quota table + counter

- ☐ Upgrade CTA on quota exhaustion
- ☐ REFINE last-query support
- ☐ One utility template approved + send path

Acceptance: end-to-end test with both an existing paid user (unlimited) and a fresh number (3-query free tier with paywall block).

PHASE 4 – SOFT LAUNCH (WEEK 5)

- ☐ Share number with 10 friendly lawyers, 1 by 1, over a week
- ☐ Monitor every conversation manually
- ☐ Fix top 5 issues / awkwardnesses
- ☐ Collect 2-3 testimonials for landing page

Acceptance: 8/10 friendly testers say "this is useful." Citation correctness ≥ 95% on real queries.

PHASE 5 – PUBLIC LAUNCH (WEEK 6+)

- ☐ Add WA number to headnote.in landing page CTA
- ☐ Add to Instagram bio + post a launch reel
- ☐ Add to distributor onboarding email
- ☐ Add to partnership proposal as a feature
- ☐ Monitor metrics weekly

• 14. Risks & mitigations

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
R1	FB Business Verification rejection	M	H	Submit clean docs first time; have alternative documents ready; budget 2 weeks for back-and-forth
R2	Display name "Headnote" rejected	L	M	Have "Headnote.in" or "Headnote Research" as fallback names
R3	LLM cost spike from abuse	L	M	Per-phone rate limit (F4); quota table; daily cost alert at ₹500
R4	Bot fabricates a citation	M	H	Use existing re-anchored pipeline; reject responses where the verifier fails; manual QA in soft launch
R5	WA number gets reported / banned	L	H	Don't send any marketing-shaped messages; respect STOP; don't broadcast to unconsented users

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
R6	Render free tier sleeps	M	M	Existing Headnote already handles cold start; webhook auto-wakes; if cold-start > 60s, upgrade to Starter (₹600/mo)
R7	User confused by free-form WA UI	M	M	Strong onboarding message; HELP keyword; soft-launch round catches this
R8	NyayAssist or competitor copies	M	M	First-mover + technical depth (re-anchoring). Citation-correctness moat is hard to copy fast
R9	Lawyer asks for sensitive personal advice	L	M	Disclaimer in PDF footer ("Not legal advice; verify before relying"); don't store opaque PII
R10	DeepSeek API downtime	L	M	Groq fallback already in pipeline; degrade gracefully with "service temporarily slow"

• 15. Open questions

Things to resolve before or during Phase 0:

1. **Headnote pricing for v1 funnel** — current paid plan price and billing cadence. Need confirmed amount for the UPGRADE CTA copy.
2. **Senior advocate co-signer** — should the bot's onboarding mention the senior advocate (credibility) or stay anonymous?
3. **PDF branding** — does the memo show Headnote logo + footer, or stay plain to allow lawyer-forwarding without giveaway?
4. **Hindi by when?** — out of scope at v1, but if early users demand it, where does it slot in? (Estimate: v2, +2 weeks)
5. **Group-chat support** — explicitly deferred. Revisit at 1,000 WAU.
6. **Document upload** — lawyer sends a PDF, asks for similar cases. Powerful but heavy. Revisit at v2.
7. **Display name** — "Headnote", "Headnote.in", "Headnote Research"? Pick before Meta submission.

• 16. Acceptance criteria for v1 GA

The bot is ready for public launch when **all** the following hold:

- ☐ FB Business verified, WABA approved, display name approved
- ☐ Webhook live on production with signature verification
- ☐ 100 representative test queries returned correct citations $\geq 95\%$

- [] P50 latency $\leq$ 30s, P95 $\leq$ 60s over 100 queries
- [] LINK flow works end-to-end with at least 3 real paid users
- [] Quota enforcement blocks the 4th free query
- [] PDF memo renders correctly across iOS and Android WhatsApp
- [] STOP keyword tested and verified silencing further messages
- [] Privacy page live at headnote.in/privacy
- [] No errors above 2% over a 72-hour soak with friendly users
- [] Total launch-month operating cost projected $\leq$ ₹500

• 17. Summary — cost recap

BUCKET	LAUNCH (MONTH 1)	SCALE (5K CONV/MO)
Setup (one-time)	₹500 (SIM)	—
Meta Cloud API	₹0 (free tier)	~₹1,300
Hosting	₹0 (Render free)	₹600
LLM	₹50–₹200	₹500–₹1,500
Storage / CDN	₹0	₹100
Total recurring	₹50–₹200	₹2,500–₹3,500
Break-even paid users (@₹999/mo)	1	4

TL;DR for the wallet: ₹500 to set up. ₹50–₹200/mo to run at launch. Pays for itself with the first converted user.

End of PRD v0.1. Update this file in place as decisions land in §15.